

ALEGATIONS, MIXTURE, SICI, SURDS AND INDICES

Directions for question 1: Select the correct alternative from the given choices.

If $x, y \neq 0$, find the value of

$$\left[\frac{\left(\frac{x}{y}\right)^n \times \left(\frac{y}{x}\right)^n + \left\{\left(\frac{y}{x}\right)^{-n} \times \left(\frac{x}{y}\right)^{-n}\right\}}{\left(\frac{x}{y}\right)^{2n} - \left(\frac{y}{x}\right)^{-2n} + \left(\frac{x+y}{x}\right)^n} \right]^{1/n}$$

- ☐ a) $(x + y)/y$
- ☐ b) $x/(x + y)$
- ☐ c) $(x + y)/x$
- ☐ d) $y/(x + y)$

Solution:

$$\frac{\left(\frac{x}{y}\right)^n \left(\frac{y}{x}\right)^n + \left(\frac{y}{x}\right)^{-n} \left(\frac{x}{y}\right)^{-n}}{\left(\frac{x}{y}\right)^{2n} - \left(\frac{y}{x}\right)^{-2n} + \left(\frac{x+y}{x}\right)^n} = \frac{1 + 1}{\left(\frac{x}{y}\right)^{2n} - \left(\frac{x}{y}\right)^{2n} + \left(\frac{x+y}{x}\right)^n} = \left(\frac{x}{x+y}\right)^n$$

The n^{th} root of this is $\frac{x}{x+y}$ Choice (B)

Directions for questions 2 and 3: Type in your answer in the input box provided below the question.

The average weight of 8 persons increases by 1.5 kg when Rom, who weighs 65 kg, is replaced by Ram. What is the weight of Ram (in kg)?

Solution:

The average weight of the original group of 8 is say x , i.e., the total is $8x$. The average weight of the new group of 8 is $x + 1.5$, i.e., the total is $8x + 12$. The extra 12 kg is the difference in the weights of the two men. The weight of Ram is $65 + 12$, i.e., 77 kg.
Ans : (77)

Directions for questions 2 and 3: Type in your answer in the input box provided below the question.

If the difference in the simple interest and the compound interest on a certain sum at 15% p.a. for two years is ₹360, find the sum (in ₹).

Solution:

The SI on P for 2 years at the rate r is $= 2rP$
The CI on P for 2 years at the rate r is $= (2r + r^2) P$
The difference $= Pr^2 = 360$
 $\therefore P = \frac{360}{r^2} = \frac{360}{(0.15)^2} = 360 \frac{(100)(100)}{15(15)} = 16000$

Ans : (16000)

Directions for questions 4 and 5: Select the correct alternative from the given choices.

If a , b and c are rational numbers and $4 + \sqrt{17}$ is one root of the quadratic equation $ax^2 + bx + c = 0$, which of the following can be the values of a , b and c ?

- ☐ a) (1, -8, 38)
- ☐ b) (1, 8, -33)
- ☐ c) (1, 8, 1)

☐ d) (1, -8, -1) (Correct Answer - ✓)

Solution:

$$ax^2 + bx + c = 0, \text{ where } a, b, c \in \mathbb{Q}$$

$4 + \sqrt{17}$ is one root of the equation.

$\therefore$ The other root has to be $4 - \sqrt{17}$.

$\therefore$ The sum of the roots $= \frac{-b}{a} = 8$ and the product $= \frac{c}{a} = -1$, i.e., $b = -8a$, $c = -a$.

$$\therefore a : b : c = 1 : -8 : -1$$

Choice (D)

Directions for questions 4 and 5: Select the correct alternative from the given choices.

$$\left[\frac{x^p}{x^q} \right]^{p-q} \div \left[\frac{x^q}{x^r} \right]^{q-r} \times \left[\frac{x^r}{x^p} \right]^{r-p} = \underline{\hspace{2cm}}.$$

☒ a) $x^{2(p-q)(p-r)}$ (Correct Answer - ✓)

☐ b) $x^{(p-q)(q-r)(r-p)}$

☐ c) $x^{2p(q-p-r)}$

☐ d) $x^{(p-q)^2 + (q-r)^2 + (r-p)^2}$

Solution:

$$\begin{aligned} \left[\frac{x^p}{x^q} \right]^{p-q} \div \left[\frac{x^q}{x^r} \right]^{q-r} \times \left[\frac{x^r}{x^p} \right]^{r-p} &= \frac{x^{(p-q)^2} x^{(r-p)^2}}{x^{(q-r)^2}} \\ \frac{x^{p^2+q^2-2pq+p^2+r^2-2pr}}{x^{q^2+r^2-2qr}} &= x^{2p^2+2qr-2pq-2pr} \\ &= x^{2(p-q)(p-r)} \end{aligned}$$

Choice (A)

Directions for question 6: Type in your answer in the input box provided below the question.

Eight kilograms of rice costing ₹16 per kg is mixed with four kilograms of rice costing ₹22 per kg. What is the average price of the mixture (in ₹)?

Your Answer: 18 ✓

Solution:

Total cost of the first mixture = ₹(8 × 16) = ₹128
Total cost of the second mixture = ₹(4 × 22) = ₹88
Average cost per kg = (128 + 88)/12 = ₹18.

Ans: (18)

Directions for question 7: Select the correct alternative from the given choices.

The simple interest on a sum of money at 20% p.a. for ten years will be _____.

- ☐ a) half the sum
- ☐ b) equal to the sum
- ☒ c) twice the sum (Your answer - ✓)
- ☐ d) thrice the sum

Solution:

The simple interest at 20% pa, for 10 years will be 200% (∵ PTR). It is twice the sum.
Choice (C)

Directions for question 8: Type in your answer in the input box provided below the question.

A solution of 66 litres contains milk and water in the ratio 7 : x. If four litres of water is added to the solution, the ratio becomes 3 : 2, find the value of x.

Your Answer: .66 ✖

Correct Answer: 4 ✔

Solution:

Initially the ratio of milk and water is 7 : x and the total quantity is 66 ℓ. Finally , the ratio is 3 : 2 and the total quantity is 70 ℓ, $\therefore$ Finally there are 42 ℓ of milk and 28 ℓ of water. Initially, there was 42 ℓ of milk and 24 ℓ of water. The ratio was 42 : 24 or 7 : 4.
 $\therefore x = 4$. Ans : (4)

Directions for questions 9 to 12: Select the correct alternative from the given choices.

The product of two consecutive multiples of 7 is fourteen more than eight times their sum. What are the two numbers?

- ☐ a) 28, 35
- ☐ b) 7, 14
- ☐ c) 21, 28
- ☐ d) 14, 21

Solution:

Let the two consecutive multiples be $7x$ and $7(x + 1)$
 $\therefore (7x) [7(x + 1)] = 8[7x + 7(x + 1)] + 14 = 112x + 70$
 $\Rightarrow 49x(x + 1) = 112x + 70$
 $\Rightarrow 7x^2 + 7x = 112x + 70 \Rightarrow 7x^2 - 105x - 70 = 0$
 $\Rightarrow (7x + 5)(x - 2) = 0 \Rightarrow x = 2$
($\because x$ is an integer) the multiples are 14 and 21.

Choice (D)

Directions for questions 9 to 12: Select the correct alternative from the given choices.

$(2b/2c)^{2-a} \times (2c/2a)^{2-b} \times (2a/2b)^{2-c}$ is equal to ____.

☐ a) $a^{b-c} \times b^{c-a} \times c^{a-b}$ (Correct Answer - ✓)

☐ b) $8abc$

☐ c) $2 \times a^{b-c} \times b^{c-a} \times c^{a-b}$

☐ d) 8

Solution:

$$\begin{aligned} & \left(\frac{2b}{2c}\right)^{2-a} \left(\frac{2c}{2a}\right)^{2-b} \left(\frac{2a}{2b}\right)^{2-c} \\ &= \frac{b^{2-a}}{c^{2-a}} \frac{c^{2-b}}{a^{2-b}} \frac{a^{2-c}}{b^{2-c}} = b^{c-a} c^{a-b} a^{b-c} \end{aligned}$$

Choice (A)

Directions for questions 9 to 12: Select the correct alternative from the given choices.

What are the values of x that satisfy the inequality $5x^2 + 48x - 77 < 0$?

☐ a) $x < -11$

☐ b) $x > 7/5$

☒ c) $-11 < x < 7/5$ (Your answer - ✓)

☐ d) Both A and B

Directions for questions 9 to 12: Select the correct alternative from the given choices.

A sum amounts to ₹56,000 in five years and ₹64,000 in six years under compound interest compounded annually. What is the rate of interest?

☐ a) 8.33% p.a.

☐ b) 12.5% p.a.

☐ c) 14.28% p.a. (Correct Answer - ✓)

☒ d) 16.67% p.a. (Your answer - ✗)

Solution:

The amount after 5 years is 56,000 and after 6 years is 64,000.

∴ The interest on 56,000 is 8000. The annual rate of interest is $= \frac{8000}{56000} = \frac{1}{7} = \frac{100}{7} \%$

= 14.28 %. The rate of interest is 14.28% pa.

Choice (C)

Directions for question 13: Type in your answer in the input box provided below the question.

The average age of a girl and her parents is 28 years. If the ratio of the present ages of the father, the mother and the girl is 7 : 6 : 1, what will be the girl's age (in years) after 2 years?

Your Answer: 8 ✓

Directions for questions 14 and 15: Select the correct alternative from the given choices.

The length of a rectangle is 5 m more than its breadth. If the area of the rectangle is 696 sq. m, what is its breadth?

☐ a) 29 m

☒ b) 24 m (Your answer - ✓)

☐ c) 36 m

☐ d) 28 m

Directions for questions 14 and 15: Select the correct alternative from the given choices.

Solve for x: $8^{\log_8 x^2} = 5^{\log_5 9} + 7^{\log_7 16}$

☐ a) 4

☐ b) 5

☐ c) -5

☐ d) Both (B) and (C)

Solution:

$$8^{\log_8 x^2} = 5^{\log_5 9} + 7^{\log_7 16}$$

We know $a^{\log_a x} = x \Rightarrow x^2 = 9 + 16$

$$x = \sqrt{25} = \pm 5$$

Choice (D)